areas in the rat's brain. This enables the scientists to manoeuvre the rat by remote control. After short training sessions, researchers have managed not only to make the rats turn left or right, but also to climb ladders, sniff around garbage piles, and do things that rats normally dislike, such as jumping from great heights. Armies and corporations show keen interest in the robo-rats, hoping they could prove useful in many tasks and situations. For example, robo-rats could help detect survivors trapped under collapsed buildings, locate bombs and booby traps, and map underground tunnels and caves.

Animal-welfare activists have voiced concern about the suffering such experiments inflict on the rats. Professor Sanjiv Talwar of the State University of New York, one of the leading robo-rat researchers, has dismissed these concerns, arguing that the rats actually enjoy the experiments. After all, explains Talwar, the rats 'work for pleasure' and when the electrodes stimulate the reward centre in their brain, 'the rat feels Nirvana'.[3]

To the best of our understanding, the rat doesn't feel that somebody else controls her, and she doesn't feel that she is being coerced to do something against her will. When Professor Talwar presses the remote control, the rat *wants* to move to the left, which is why she moves to the left. When the professor presses another switch, the rat *wants* to climb a ladder, which is why she climbs the ladder. After all, the rat's desires are nothing but a pattern of firing neurons. What does it matter whether the neurons are firing because they are stimulated by other neurons, or because they are stimulated by transplanted electrodes connected to Professor Talwar's remote control? If you asked the rat about it, she might well have told you, 'Sure I have free will! Look, I want to turn left – and I turn left. I want to climb a ladder – and I climb a ladder. Doesn't that prove that I have free will?'

Experiments performed on *Homo sapiens* indicate that like rats humans too can be manipulated, and that it is possible to create or annihilate even complex feelings such as love, anger, fear and depression by stimulating the right spots in the human brain. The US military has recently initiated experiments on implanting computer chips in people's brains, hoping to use this method to treat soldiers suffering from post-traumatic stress disorder.[4] In